Retail Sales Analysis

Model Answers • Business Questions • Insights • Recommendations & Action Plan

Purpose: This document provides one defensible model answer for the Retail Sales Analysis graduation project. Students may reach different conclusions if they document and justify a different cleaning policy.

1. Data Cleaning and Preparation

• **Raw dataset:** 1,515 rows and 13 columns.

• **Duplicate transactions:** 15 duplicate Transaction_ID rows were removed, leaving 1,500 unique transaction IDs.

• **Invalid numeric values:** Rows with non-positive unit price, discount outside the accepted 0%–50% range (including missing discounts), non-positive sales, or invalid cost/profit logic were excluded from the main KPI analysis.

• **Extreme sales anomalies:** Six transactions had reported sales above EGP 5,000 despite small quantities and unit prices (for example, a two-unit item priced at EGP 132 with reported sales of EGP 25,000). These were excluded from the primary KPI view and should be checked against source receipts rather than silently overwritten.

• **Final analysis set:** 1,454 transactions. Missing payment method and customer type values were retained for numeric sales/profit calculations, but should be shown as “Unknown” in segment charts.

• **Derived metrics:** Profit Margin = Total Profit ÷ Total Sales × 100. Use weighted margin from summed values, not the simple average of row-level margins.

2. Executive Summary

Metric	Model result
Transactions analyzed	1,454
Net sales	EGP 690,269.37
Recorded profit	EGP 164,109.78
Overall profit margin	23.77%
Average sales per transaction	EGP 474.74
Top sales branch	Main Branch — EGP 306,300.87
Top sales category	Cookware — EGP 343,808.99
Highest-margin category	Cookware — 24.80%

Important interpretation: these are synthetic project data and the profit values are the dataset’s recorded Profit_EGP values. Treat them as analysis outputs, not audited financial statements.

3. Model Answers to Business Questions

Q1. Which branch generates the highest sales and profit?

The **Main Branch** has the highest sales (EGP 306,300.87) and highest total profit (EGP 73,231.68), across 608 transactions. The University Branch generates EGP 233,656.03 sales and EGP 54,933.73 profit. The Market Branch has the lowest total sales (EGP 150,312.47) but a similar weighted margin to Main Branch (about 23.91%).

Business meaning: Main Branch is the largest contributor by volume and total profit. Compare branch footfall, inventory availability, operating costs, and local demand before assuming it is the most efficient branch; this dataset does not include rent or staffing costs.

Q2. Which product categories perform best?

Category	Sales (EGP)	Profit (EGP)	Margin
Cookware	343,808.99	85,270.63	24.80%
Drinkware	143,668.90	34,759.91	24.19%
Kitchen Storage	132,577.24	29,026.07	21.89%
Cleaning Tools	70,214.24	15,053.17	21.44%

Cookware leads in sales, profit, and margin. Drinkware is the second-largest category by sales and has a strong margin. Cleaning Tools has the lowest category sales and margin, so review pricing, sourcing costs, product mix, and whether it attracts add-on purchases before reducing its range.

Q3. Which individual products are the strongest and weakest performers?

Product	Sales (EGP)	Profit (EGP)	Margin
Cooking Pot	159,066.37	37,269.37	23.43%
Frying Pan	84,386.60	22,138.04	26.23%
Thermal Flask	60,718.40	17,141.60	28.23%
Kitchen Knife Set	60,189.47	17,974.63	29.86%
Lunch Box	30,382.80	5,929.80	19.52%
Dish Brush	8,139.84	1,307.44	16.06%
Spray Bottle	6,550.26	1,444.74	22.06%

Cooking Pot is the largest product by sales and profit. Kitchen Knife Set has the highest listed product margin (29.86%), while Thermal Flask also combines strong sales and margin. Lunch Box sells frequently but has a lower margin than the leading products. Dish Brush has a low margin and low profit contribution; investigate purchase cost, selling price, supplier terms, and discounting before deciding whether to reprice or bundle it.

Q4. Do discounts affect profit margin?

Discount	Transactions	Sales (EGP)	Profit (EGP)	Margin
0%	644	333,471.00	96,069.09	28.81%
5%	208	99,883.70	25,283.24	25.31%
10%	223	96,669.26	20,092.11	20.78%
15%	190	77,154.01	12,768.36	16.55%
20%	189	83,091.40	9,896.98	11.91%

The observed margin declines as discount levels rise: from 28.81% with no discount to 11.91% at 20% discount. This is a strong warning sign that deep discounts can erode profitability. However, this is observational data: products offered larger discounts may differ from products sold without discounts. A controlled test or product-level comparison is needed before claiming discounts alone caused the decline.

Q5. Which customer types and payment methods contribute most?

Walk-in customers contribute the largest sales (EGP 323,330.76) and profit (EGP 76,273.79). Returning customers contribute EGP 163,204.01 in sales and EGP 39,503.11 in profit. Business Customers have the smallest sales total (EGP 60,611.63) but the highest aggregate margin among customer types (approximately 25.13%). Cash is the most-used payment method and has the highest total profit; Mobile Wallet has the highest payment-method margin among the listed methods (about 25.23%). Segment results should be interpreted alongside transaction count and missing customer/payment labels.

Q6. How do sales and profit change over time?

April has the highest monthly sales in the cleaned analysis set (EGP 89,807.82) and highest monthly profit (EGP 21,838.04). January has EGP 74,446.94 sales and EGP 17,828.35 profit. September has lower sales (EGP 67,164.12) and 137 analyzed transactions, but it may be an incomplete month in the source data. Confirm date coverage before describing September as a genuine decline. The data covers January–September 2026, so it is not sufficient to claim a full-year seasonal pattern.

4. Key Business Insights

Cookware is the core category. It generates about half of cleaned sales and more than half of recorded profit, with a 24.80% margin. Keep availability high and monitor stockouts for the strongest cookware products.

Main Branch leads in scale, not necessarily in efficiency. It contributes the highest sales and profit, while Market Branch has a similar margin. A full profitability comparison requires rent, payroll, utilities, and branch-level operating costs.

Discount depth is associated with lower margins. Margin falls from 28.81% at 0% discount to 11.91% at 20%. Discount rules should be linked to minimum margin and product-level economics.

Product performance varies significantly within categories. Cooking Pot leads sales/profit, while Kitchen Knife Set has the highest listed product margin. Lunch Box and Dish Brush deserve margin reviews, not automatic delisting.

Data quality materially changes conclusions. Duplicate IDs, invalid prices/discounts, and six implausibly high sales values can distort rankings and total revenue. Cleaning decisions must be documented and exceptions reviewed.

Customer segments suggest retention and B2B opportunities. Walk-ins contribute the largest total sales, while Business Customers have a relatively strong margin. Test loyalty offers for returning customers and targeted bundles/volume terms for business buyers.

5. Recommendations and Actions

Priority	Recommended action	Measure of success
High	Audit the six extreme sales records against invoices/POS records; preserve flagged data and document resolved corrections.	All flagged records reviewed and audit log retained.
High	Set product-level minimum margin thresholds and require approval for discounts that push margin below them.	Weighted margin and profit per transaction; share of sales below threshold.
High	Prioritize replenishment for Cooking Pot, Frying Pan, Thermal Flask, and Kitchen Knife Set based on sales, margin, and supplier lead time.	Stockout rate, sell-through, and on-time supplier delivery.
Medium	Review Lunch Box, Dish Brush, and Cleaning Tools costs/prices; negotiate with suppliers, bundle, or reduce promotion for low-margin products.	Margin by SKU, units sold, and contribution profit.
Medium	Create a branch scorecard combining sales, profit, margin, transaction count, and operating expenses.	Branch-level profit after controllable costs.
Medium	Run a controlled discount test: compare matched products/branches with different discount levels.	Incremental units sold and profit versus a no-discount/control group.
Medium	Build a simple dashboard with filters for date, branch, category, product, customer type, and payment method.	Weekly use by managers; consistent KPI definitions.
Low	Test a loyalty campaign for returning customers and tailored volume bundles for Business Customers.	Repeat purchase rate, average order value, and incremental profit.

6. Suggested 30-Day Action Plan

Timeline	Action	Owner / output
Week 1	Validate duplicates, missing fields, impossible prices/discounts, and six extreme sales records.	Analyst + store manager; signed-off clean dataset and data-quality log.
Week 2	Review SKU economics and supplier costs; establish minimum margin rules.	Category manager; SKU margin sheet and discount guardrails.
Week 3	Review branch performance including operating costs; identify stockouts and slow movers.	Branch manager; branch scorecard and inventory actions.
Week 4	Launch one controlled promotion test and publish the KPI dashboard.	Marketing + analyst; test design, dashboard, and follow-up date.

7. Suggested Dashboard KPIs

• Total Sales (EGP), Total Recorded Profit (EGP), Weighted Profit Margin (%), Transaction Count, Units Sold, and Average Sales per Transaction.

• Sales and profit trends by month; branch comparison; category and product ranking.

• Discount level vs weighted margin; product-level margin table with conditional formatting.

• Customer type and payment method contribution, including an explicit Unknown category for missing labels.

• Data-quality indicators: duplicate IDs, invalid price/discount records, extreme sales outliers, and missing segment values.

8. Trainer Notes and Limitations

• The dataset is synthetic and includes intentional data-quality issues. It is suitable for teaching data cleaning and analysis, not for making real financial decisions.

• The main figures above exclude duplicate transaction IDs, invalid numeric records, and six extreme reported sales values. If students use a different defensible cleaning policy, their totals may differ.

• The recorded Profit_EGP values were summed as provided. Students should check whether Sales_EGP – Cost_EGP equals Profit_EGP and investigate inconsistencies before presenting the results as accounting profit.

• Do not infer that discounts cause lower margin from a simple grouped comparison; product mix and promotion selection may confound the relationship.

• September may be incomplete. Verify date coverage before comparing it with full months.

• Branch profitability cannot be fully assessed without rent, payroll, utilities, and other operating expenses.

Suggested final presentation: State the business problem, explain cleaning decisions, present 4–6 KPIs, show the strongest and weakest products/categories, summarize 3–5 evidence-based insights, and finish with prioritized actions and measurable success metrics.